Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 1 of 8 · Question 5 part b.iii · 2 marks

2023 JUN · Paper 01 · extract 1 of 8 · find part b.iii in this mark scheme extract

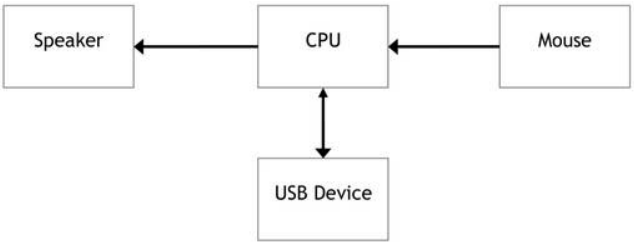
<table>
<tr>
<td>5(a)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Req.</th><th>Example</th></tr>
<tr><td>Input</td><td>1. Enter the direction the bee will face (1)
2. Enter the number of steps that the bee will move (1)</td></tr>
<tr><td>Process</td><td>Calculate the path the bee will move along to its new position</td></tr>
<tr><td>Output</td><td>3. The bee appears in its new position (1)
4. The bee flies to its new position (1)
5. The bee faces its new direction (1)</td></tr>
</table>
</td>
<td>Allow direct instructions as input e.g. ‘Turn East’ and consequences for output e.g. ‘Bee turns to face East’ if clear.

The input/output do not have to be directly linked. E.g. Input Turn West, Output Has Moved 20 steps</td>
<td>2</td>
</tr>
</table>

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 1 of 8 · Question 5 part b.iii · 2 marks

2023 JUN · Paper 01 · extract 2 of 8 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5b(i)	Award one mark for each: • Line between Mouse and CPU with arrow at CPU end only (1) • Line between CPU and Speaker with arrow at Speaker end only (1) • Line between CPU and USB Device with arrows on both ends (1) 	Allow two independent back/forth arrows between CPU and USB in place of a single bi-directional arrow. Ignore any additional lines.	3

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 1 of 8 · Question 5 part b.iii · 2 marks

2023 JUN · Paper 01 · extract 3 of 8 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(ii)	**The only correct answer is B** ***A*** *is not correct because a* ***firewall*** *is an application that is loaded after the device is booted* ***C*** *is not correct because* ***the operating system*** *is loaded after the device is booted* ***D*** *is not correct because* ***the user interface*** *is loaded after the device is booted*		1

Question Number	Answer	Additional Guidance	Mark
5(b)(iii)	Award up to **two** marks for a linked explanation, such as: • Cache makes up for the difference in speed of two devices (1) because it is very high-speed memory (1) • Cache is high-speed memory (1) which acts as a buffer between a faster and a slower device (1) • It speeds up processing (1) because active/commonly used instructions and data are stored nearer to the CPU / the CPU doesn't have to wait while instructions/data are fetched from RAM (1)		2

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 1 of 8 · Question 5 part b.iii · 2 marks

2023 JUN · Paper 01 · extract 4 of 8 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)	Award up to **two** marks for a linked description, such as: • Uses a sensor (1) to check the water level in the machine (1) • Uses a thermometer (1) to check the water temperature (1) • Uses a timer (1) to control the time required for the wash cycle/rinse/drain/spin (1) • Uses a motor (1) to rotate the drum (1)	The first mark is for identification and the second is for the linked description. It is possible to gain one mark for a description without identifying the sensor/actuator.	2

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 1 of 8 · Question 5 part b.iii · 2 marks

2023 JUN · Paper 01 · extract 5 of 8 · find part b.iii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(i)</td><td>Award one mark for each two correct rows.

<table>
<tr><th>A</th><th>B</th><th>C</th><th>A AND C</th><th>(A AND C) OR B</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>0</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr>
</table>
</td><td></td><td>3</td></tr>
</table>

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 1 of 8 · Question 5 part b.iii · 2 marks

2023 JUN · Paper 01 · extract 6 of 8 · find part b.iii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>5(d)(ii)</td><td>Award one mark for each of:

• M OR P (1)
• NOT C (1)
• <expr1> AND <expr2> (1)

Example statements:

(M OR P) AND (NOT C)
(M OR P) AND NOT C
(M AND NOT C) OR (P AND NOT C)
M AND NOT C OR P AND NOT C

For information only:

<table>
<tr><th>M</th><th>P</th><th>C</th><th>M OR P</th><th>NOT C</th><th>(M OR P) AND (NOT C)</th></tr>
<tr><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>0</td></tr>
<tr><td>0</td><td>0</td><td>1</td><td>0</td><td>0</td><td>0</td></tr>
<tr><td>0</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>0</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
<tr><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td></tr>
</table>
</td><td>Third bullet awarded even if first two are not

Order of precedence is NOT, AND, OR

Check brackets carefully, if provided</td><td>3</td></tr>
</table>

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 1 of 8 · Question 5 part b.iii · 2 marks

2023 JUN · Paper 01 · extract 7 of 8 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(e)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. The operating system uses a scheduler to control processes (1) 2. The operating system holds processes in a queue (1) 3. Some processes may be given higher priorities than others (1) 4. Each process gains accesses to the CPU for a short time / time slice to execute (1) 5. Processes are swapped to/from (the queue / CPU)	Do not award responses about memory management, as the question is about sharing the CPU only	2

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 1 of 8 · Question 5 part b.iii · 2 marks

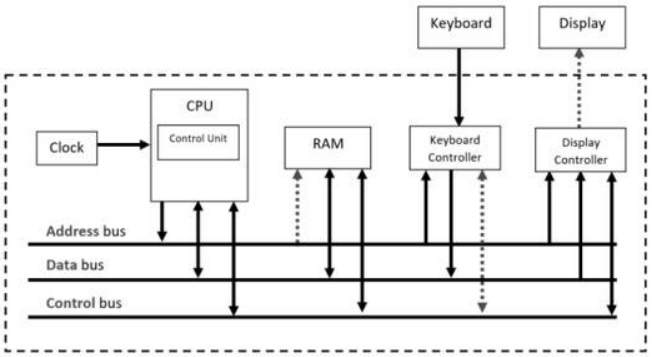
2023 JUN · Paper 01 · extract 8 of 8 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 2 of 8 · Question 3 part b · 2 marks

2020 NOV · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

3(a)	 • **Address bus** labelled correctly (1) • **Data bus** labelled correctly (1) • **Control bus** labelled correctly (1) • Arrow from address bus to RAM (1) • Bi-Directional arrows from control bus to keyboard controller (1) • Arrow points from display controller to display (1)		6

Question Number	Answer	Additional Guidance	Mark
3(b)	Award **two** marks for a linked explanation such as: The cache stores frequently used data/instructions (1) so the processor does not have to wait because cache is checked before main memory / because RAM is further away from the processor (1). It speeds up processing (1) because cache is a faster type of memory (1).		2

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 2 of 8 · Question 3 part b · 2 marks

2020 NOV · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>3(c)</td><td>
<table>
<tr><th>CPU1</th><th>CPU2</th></tr>
<tr><td>5 * 2 = 10</td><td>4 * 3 = 12 (1)</td></tr>
<tr><td>10 + 12 = 22 (1)</td><td>24 / 4 = 6 (1)</td></tr>
<tr><td>22 - 6 = 16 (1)</td><td></td></tr>
</table>
</td><td>The second row expressions can be in either column

22 – 6 = 16
May appear in either CPU in the final row</td><td>4</td></tr>
</table>

Question Number	Answer	Additional Guidance	Mark
3(d)	Any four from: • The OS checks whether sufficient space is available on the storage (media) for the file (1). • The file is broken into blocks (1) • The blocks are stored in spaces that are large enough (1) • OS looks for (fat/ntfs)/ reserves (linux) (groups of) contiguous blocks (1) • Blocks can reside anywhere on the storage (1). • The OS updates the file allocation table (with the start location and sequence number of each block) (1) • OS sets hard links to files (1) ntfs • OS updates journal/MFT (1) ntfs/linux • Metadata about the file such as read/write permissions, date created and last accessed is separately stored (1)		4

Match each answer number to the type practice PDF. Original exam material © Pearson Education Ltd.

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 2 of 8 · Question 3 part b · 2 marks

2020 NOV · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 3 of 8 · Question 3 part c · 2 marks

2021 JUN · Paper 01 · extract 1 of 2 · find part c in this mark scheme extract

Number		Guidance	
3(a)	Award **two** marks for an explanation that includes hardware **and** software. • The laptop is/contains the hardware (1) and the software allows the user to interact with it (1) • Hardware gives input and output devices (1) /memory, storage, cpu, processing ability • Software allows instructions to be passed to the hardware (1) so that it can carry out tasks (1) • Software is the set of instructions (1) that controls the hardware (1)		2

Question Number	Answer	Additional Guidance	Mark
3(b)(i)	Award **two** marks from: • Lighter/thinner (1) • Less heat generated (1) • Faster access speed / faster booting of operating system / faster data transfer/read/write speeds (1) • Silent operation (1) • Less power required / longer battery life (1) • More robust (due to no moving parts) (1)		2
3(b)(ii)	Award **two** marks for a linked explanation such as: • Flash memory is non-volatile (1) whereas RAM is volatile (1) • Flash memory persists even when power is lost (1) whereas RAM clears its data whenever power is lost (1) • Flash memory is used for storage (1) whereas RAM is used in performing operations on data taken from storage (1)		2

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 3 of 8 · Question 3 part c · 2 marks

2021 JUN · Paper 01 · extract 2 of 2 · find part c in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
3(c)	Award **two** marks for a linked explanation such as: • Cache memory stores active/commonly used instructions / cache is a temporary storage (1) and is used to speed up processing. (1) stop data transfer bottlenecks between RAM and CPU (1)		2

Question Number	Answer	Additional Guidance	Mark
3(d)	Award **one** mark for two correct rows. Award **two** marks for four correct rows.		2

Device	Input	Output	Both
Headset			✓
Mouse	✓		
Printer		✓	
Touch screen			✓

Question Number	Answer	Additional Guidance	Mark
3(e)	Award **two** marks from: • Antivirus (1) • Anti-spyware (1) • (Software) firewall (1)	Do not award tradenames	2

Question Number	Answer	Additional Guidance	Mark

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 4 of 8 · Question 5 part b.iii · 1 mark

2021 NOV · Paper 01 · extract 1 of 4 · find part b.iii in this mark scheme extract

5(a)	Award **one** mark for each correct component: **Component** / **Letter** Control unit — B Output device — G Registers — C Main memory — H Cache — F Input device — E Arithmetic/logic unit (ALU) — D Clock — A		5

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 4 of 8 · Question 5 part b.iii · 1 mark

2021 NOV · Paper 01 · extract 2 of 4 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(b)(i)	Award **one** mark each to a maximum of **two** marks for: • stores a memory address (for the next instruction) (1) • memory address incremented/changed during the fetch process (1)		2
5(b)(ii)	**The only correct answer is B** A *is not correct because* there is no such signal C *is not correct because* as there is no such signal D *is not correct because* as that does not take place during the fetch stage		1
5(b)(iii)	Award **one** mark from: • processes more instructions in the same amount of time (1) • more cycles can be performed in the same amount of time (1) • fetch-execute-decode cycle runs faster (1)		1

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 4 of 8 · Question 5 part b.iii · 1 mark

2021 NOV · Paper 01 · extract 3 of 4 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
5(c)(i)	Award **one** mark from: • a system designed for one/a few specific functions/task(s) (1) • has both hardware and software (1) • has integrated memory (1) • has no or minimal user interface (1) • is power efficient/low power consumption (1) • its functionality cannot be changed/upgraded by users (1) • often use sensors and actuators to interact with the external environment (1) • functions in real time (1)		2
5(c)(ii)	Award up to **two** marks for a linked explanation such as: • Performance / number of cores / power consumption needs to be adequate for the task (1) because any more would waste money / power (1) • Size of cache can be small or non-existent (1) because there is no need to store frequently used instructions (1) • The RAM can be limited/very small (1) because it is only holding a limited number of instructions (1) • Only needs a small bus width (1) because the output signals are serial binary (1)		2
		Total	**13**

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 4 of 8 · Question 5 part b.iii · 1 mark

2021 NOV · Paper 01 · extract 4 of 4 · find part b.iii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 5 of 8 · Question 1 part c.i · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part c.i in this mark scheme extract

1(a)(i)	Award **one** mark from: • An action/task that is carried out (by the CPU) (1) • An operation/code that is/will be executed (by the CPU) (1) • Something that tells the CPU/processor/device what to do (1)	Allow examples	1
1 (a)(ii)	Award **one** mark from: • A pointer/reference to/identification of a location in memory (1) • A pointer/reference to/location where data will be accessed from/stored (to)(1)		1

Question Number	Answer	Additional Guidance	Mark
1(b)	**The only correct answer is D** *A is not correct because the address bus carries the address of the memory location* *B is not correct because the data bus carries but does not store data* *C is not correct because the control unit sends signals but does not handle the data*		1

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award **one** mark from: • More instructions can be carried out per second (1) • Processes run faster (1) • Programs executed faster (1) • Can run more complex programs (1)		1
1(c)(ii)	Award **one** mark from: • The CPU/computer could overheat (1) • More cooling required (1) • Needs more power (1) • CPU could become unstable/crash / its lifespan could be shortened (1)		1

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 5 of 8 · Question 1 part c.i · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(d)	**The only correct answer is D** *A is not correct because it describes a method of accessing data from a secondary storage device* *B is not correct because it describes the multi-agent computational model* *C is not correct because it describes the parallel computational model*		1

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 5 of 8 · Question 1 part c.i · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part c.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(e)(i)</td><td>Award one mark from:

• When fast execution of the program is essential/gives faster execution (1)
• When writing code to control a device / directly access the hardware (1)
• To make efficient use of a device's limited storage/memory/power (1)
• When writing code for a particular architecture/hardware/embedded system (1)</td><td></td><td>1</td></tr>
<tr><td>1(e)(ii)</td><td>Award one mark for:

• To translate/convert assembly language into machine/object code (1)</td><td></td><td>1</td></tr>
<tr><td>1(e)(iii)</td><td>Award one mark for any two correct rows (1)
Award two marks for any four correct rows (2)
Award three marks for all five correct rows (3)

<table><tr><th>Description</th><th>Compiler</th><th>Interpreter</th></tr><tr><td>Translates the program each time it is executed</td><td></td><td>☐</td></tr><tr><td>Produces permanent object code</td><td>☐</td><td></td></tr><tr><td>Translates line by line</td><td></td><td>☐</td></tr><tr><td>Translates the whole program before it is run</td><td>☐</td><td></td></tr><tr><td>Generates a list of errors once the complete program has been translated</td><td>☐</td><td></td></tr></table></td><td>Row can only be correct if there is only one tick.</td><td>3</td></tr>
<tr><td colspan="3">Total for question 1</td><td>11</td></tr>
</table>

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 5 of 8 · Question 1 part c.i · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part c.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 6 of 8 · Question 1 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 1 of 4 · find part c.ii in this mark scheme extract

1(a)(i)	Award **one** mark from: • An action/task that is carried out (by the CPU) (1) • An operation/code that is/will be executed (by the CPU) (1) • Something that tells the CPU/processor/device what to do (1)	Allow examples	1
1 (a)(ii)	Award **one** mark from: • A pointer/reference to/identification of a location in memory (1) • A pointer/reference to/location where data will be accessed from/stored (to)(1)		1

Question Number	Answer	Additional Guidance	Mark
1(b)	**The only correct answer is D** *A is not correct because the address bus carries the address of the memory location* *B is not correct because the data bus carries but does not store data* *C is not correct because the control unit sends signals but does not handle the data*		1

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award **one** mark from: • More instructions can be carried out per second (1) • Processes run faster (1) • Programs executed faster (1) • Can run more complex programs (1)		1
1(c)(ii)	Award **one** mark from: • The CPU/computer could overheat (1) • More cooling required (1) • Needs more power (1) • CPU could become unstable/crash / its lifespan could be shortened (1)		1

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 6 of 8 · Question 1 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(d)	**The only correct answer is D** *A is not correct because it describes a method of accessing data from a secondary storage device* *B is not correct because it describes the multi-agent computational model* *C is not correct because it describes the parallel computational model*		1

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 6 of 8 · Question 1 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part c.ii in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(e)(i)</td><td>Award one mark from:

• When fast execution of the program is essential/gives faster execution (1)
• When writing code to control a device / directly access the hardware (1)
• To make efficient use of a device's limited storage/memory/power (1)
• When writing code for a particular architecture/hardware/embedded system (1)</td><td></td><td>1</td></tr>
<tr><td>1(e)(ii)</td><td>Award one mark for:

• To translate/convert assembly language into machine/object code (1)</td><td></td><td>1</td></tr>
<tr><td>1(e)(iii)</td><td>Award one mark for any two correct rows (1)
Award two marks for any four correct rows (2)
Award three marks for all five correct rows (3)

<table><tr><th>Description</th><th>Compiler</th><th>Interpreter</th></tr><tr><td>Translates the program each time it is executed</td><td></td><td>☐</td></tr><tr><td>Produces permanent object code</td><td>☐</td><td></td></tr><tr><td>Translates line by line</td><td></td><td>☐</td></tr><tr><td>Translates the whole program before it is run</td><td>☐</td><td></td></tr><tr><td>Generates a list of errors once the complete program has been translated</td><td>☐</td><td></td></tr></table></td><td>Row can only be correct if there is only one tick.</td><td>3</td></tr>
<tr><td colspan="3">Total for question 1</td><td>11</td></tr>
</table>

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 6 of 8 · Question 1 part c.ii · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part c.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 7 of 8 · Question 5 part b · 2 marks

2024 JUN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

5(a)	Award **one** mark for each correct column: Translator Characteristic / Assembler / Interpreter / Compiler Does **not** have to be present in memory to execute the translated code. — ✓ — — ✓ Translates high level language source code. — — ✓ ✓ Executes high level code until either the program has finished, or a syntax error is encountered. — — ✓ — Translates low level language source code into machine code. — ✓ — —		3
5(b)	Award **two** marks for a linked explanation such as: • Fewer instructions are executed per second (1) so gameplay will not appear to execute as smoothly / graphics lag / fewer frames per second / can only handle less complex graphics (1) • Consoles will execute more instructions per second / have faster processors / GPUs (1) so gameplay will not be realistic / mirrored / fair comparison (1)	One mark can be awarded processing speed implication, but the second mark is dependent on having a gameplay impact being identified. e.g. "A lower clock speed means processing will take longer" is generic for one mark. Low performance on its own is not enough.	2

16

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 7 of 8 · Question 5 part b · 2 marks

2024 JUN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

<table>
<tr>
<td>5(c)</td>
<td>Award one mark for each correct cell:
<table>
<tr><th>Issue</th><th>Utility software</th></tr>
<tr><td>Her word processor cannot read all of the files because it is corrupt.</td><td>File repair/recovery</td></tr>
<tr><td>She has a JPEG image and needs to send it to a publisher who can only read PNG image files.</td><td>Converting</td></tr>
<tr><td>There has been a flood and files on a server have been damaged, but Anna needs to access her data.</td><td>Backup/restore</td></tr>
<tr><td>Anna's hard drive is running very slowly when she tries to access a file.</td><td>Defragmentation/defrag</td></tr>
</table>
</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>5(d)</td>
<td>Award one mark for:
• Lossless (1)

Award one mark for any of the following up to a maximum of two marks:
• No data will be lost / the process is reversable (1)
• The manual will still be readable (1)
• Removal of data would change the meaning of text (1)
• Proofreader will be able to edit the text (1)</td>
<td>Allow converse if candidate gives reasons why lossy would not be used for justification

Allow RLE as a valid lossless compression method.</td>
<td>3</td>
</tr>
</table>

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 7 of 8 · Question 5 part b · 2 marks

2024 JUN · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

5(e)	Award **one** mark for each of: Method 1 • (C AND NOT D) (1) • (NOT C AND D) (1) • <expr1> OR <expr2> (1) Method 2 • (C OR D) (1) • NOT (C AND D) (1) • <expr1> AND <expr2> (1) Example statements: (C AND NOT D) OR (NOT C AND D) (C OR D) AND NOT (C AND D)	Order of precedence is NOT, AND, OR Note other equivalent solutions MAX 2 if all terms correct apart from missing brackets that are required. Do not credit circuit diagrams.	3
		Total for question 5	**15**

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 7 of 8 · Question 5 part b · 2 marks

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Question Number	Answer	Additional Guidance	Mark

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 8 of 8 · Question 2 part c.i · 3 marks

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2(a)	Award **one** mark for any of the following up to a maximum of **two** marks: • (Touch)screen (1) • Speaker (1) • Vibration (motor) (1) • Flashlight/LED light/torch (1) • NFC chip (1)	**Accept:** • display for screen **Do not accept:** • monitor • printer • headset • other peripheral devices	**2**
Question	Answer	Additional Guidance	Mark
2(b)(i)	**The only correct answer is D** *A is not correct because cache is located closer to the CPU* *B is not correct because random-access memory is a physical card on the motherboard* *C is not correct because read-only memory cannot be written to*		**1**
Question	Answer	Additional Guidance	Mark
2(b)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • Often held/stored remotely/stored on the internet (1) • Held/managed by a third party (1) • Requires internet connectivity to access (1)		**2**

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Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 8 of 8 · Question 2 part c.i · 3 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part c.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark
2(c)(i)	Award **one** mark for any of the following up to a maximum of **three** marks: • Clock speed/number of cycles per second (1) • Number of processor cores (1) • Size of cache (memory) (1) • Type of cache (1) • Width of data bus/word length (1) • Width of address bus (1) • Whether the CPU is overclocked or not (1) • Pipelining architecture (1) • Thermal throttling / cooling system (reducing clock speed) (1)	**Accept:** • units of clock speed e.g. GHz for MP1 • width of bus for either MP5 or MP6 but not both **Do not accept:** cache without clarification of size or type	3
Question	Answer	Additional Guidance	Mark
2(c)(ii)	Award up to **two** marks for a linked description such as: • The CPU places an address on to the address bus (1) to specify the location in main memory (1) • (The address bus) contains the location in RAM (1) to read from/write to (1) • The value in the memory address register (MAR) is placed on the address bus (1) to access the correct location in RAM (1) • The address bus carries the memory location (1) of the current/next instruction (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
Question	Answer	Additional Guidance	Mark
2(d)	Award **one** mark for any of the following up to a maximum of **two** marks: • Sub-atomic particles (1) • Qubit(s) (1) • Entanglement (1) behaviour of a pair of qubits can influence each other • Quantum parallelism (1) - allowing a quantum computer to evaluate multiple computational paths simultaneously	**Do not accept:** 'Qubits can exist in a state of 1 and 0 at the same time' as this is the same as 'superposition' given in the question	2
		Total for question 2	**12**

Type MS · What makes a CPU faster: clock speed, cores and cache

Answer 8 of 8 · Question 2 part c.i · 3 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part c.i in this mark scheme extract

Question	Answer	Additional Guidance	Mark